Matthew Abate, Ph.D.

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Biographical Sketch

I am an applied mathematician and researcher, living in Jersey City, New Jersey.

From 2017 to Fall 2022, I was a Ph.D. student and academic researcher at Georgia Tech. My research proposed new math-theoretic tools and algorithms studying systems of differential equations with disturbances.

I'm currently employed with TextQL Inc., an New York based AI startup that I joined in 2024. My primary role at TextQL is to manage our internal data systems and our internal deployment of AI agents.

Outside of work, I enjoy solving puzzles and studying sequences and patterns. I hope to one day find a solution to the Collatz Conjecture.

Education

Georgia Institute of Technology - Atlanta, GA — *August 2022*
Doctor of Philosophy in Robotics (Advisors: Samuel Coogan and Eric Feron)
Dissertation Title: Mixed Monotonicity for Efficient Reachability with Applications to Robust Safe Autonomy (site, pdf)

Georgia Institute of Technology - Atlanta, GA — *December 2021*
Master of Science in Electrical and Computer Engineering

Georgia Institute of Technology - Atlanta, GA — *December 2018*
Master of Science in Mechanical Engineering

Thayer School of Engineering at Dartmouth - Hanover, NH — *June 2017*
Bachelor of Engineering: Concentration in Mechanical Engineering

Dartmouth College - Hanover, NH — *June 2017*
Bachelor of Arts: Double Major in Engineering Sciences, and Music

Technical Work as an Entrepreneur

Determinism Library — *March 2024*
Algorithms for modeling nondeterministic processes such as the input-output behavior of language models. Implemented as a Python library with tools from statistics and combinatorics.

Webscraper-Writer Agent — *October 2023*
AI Agent for writing webscraping code, aimed at gathering data from websites with large sitemaps. Implemented in Python using Langchain, AutoGPT and the OpenAI api.

Multi-Camera Fusion Library — *December 2022*
Algorithms for merging spatial data from many disparate camera feeds. Implemented as a Python library with and tools from linear algebra, matrix calculus and numerical optimization.

Spatial Perception Library — *August 2022*
Computer vision algorithms for reconstructing 3D scenes from a monocular camera. Implemented as a Python library with tools from differential equations, interval arithmetic, and contraction theory.

Articles in Journals or Magazines

[J9] **M. Abate**, M. Mote, M. Dor, C. Klett, S. Phillips, K. Lang, P. Tsiotras, E. Feron, and S. Coogan,"Run Time Assurance for Spacecraft Attitude Control Under Nondeterministic Assumptions," *IEEE Transactions on Control Systems Technology*, pp. 1-12, Dec. 2023. DOI: 10.1109/TCST.2023.3340624.

[J8] H. Abdelraouf, E. Feron, C. Klett, **M. Abate**, and S. Coogan, "Hierarchy of Quadratic Lyapunov Functions for Linear Time-Varying and Related Systems," *Journal of Guidance, Control, and Dynamics*, pp. 1-12, Dec. 2023. DOI: 10.2514/1.G006658.

[J7] K. Hobbs, M. Mote, **M. Abate**, S. Coogan, and E. Feron, "Run Time Assurance for Safety-Critical Systems: An Introduction to Safety Filtering Approaches for Complex Control Systems," in *IEEE Control Systems Magazine (CSM)*, 2022. DOI: 10.1109/MCS.2023.3234380.

[J6] **M. Abate** and S. Coogan, "Robustly Forward Invariant Sets for Mixed-Monotone Systems," in *IEEE Transactions on Automatic Control (TAC)*, vol. 67, no. 9, pp. 4947-4954, Sept. 2022. DOI: 10.1109/TAC.2022.3170863.

[J5] **M. Abate** and S. Coogan, "Decomposition Functions for Interconnected Mixed Monotone Systems," in *IEEE Control Systems Letters (L-CSS)*, vol. 6, pp. 2120-2125, Jan. 2022. DOI: 10.1109/LCSYS.2021.3136652.

[J4] M. Srinivasan, **M. Abate**, G. Nilsson and S. Coogan, "Extent-Compatible Control Barrier Functions," in *Systems & Control Letters*, vol. 150, Mar. 2021. DOI: 10.1016/j.sysconle.2021.104895.

[J3] **M. Abate**, W. Stuckey, L. Lerner, E. Feron and S. Coogan, "Memory-Loss Resilient Controller Design for Temporal Logic Constraints," in *Cyber-Physical Systems*, vol. 7, no. 4, pp. 221-242, Oct. 2020. DOI: 10.1080/23335777.2020.1837248.

[J2] **M. Abate**, M. Dutreix and S. Coogan, "Tight Decomposition Functions for Continuous-Time Mixed-Monotone Systems with Disturbances," in *IEEE Control Systems Letters (L-CSS)*, vol. 5, no. 1, pp. 139-144, Jan. 2021. DOI: 10.1109/LCSYS.2020.3001085.

[J1] D. Wang, N. Hu, S. Huang, A. M. Nasab, K. Yang, **M. Abate**, X. Yu, L. Tan, W. Shan, Z. Chen, "Buckling and post-buckling of an elastic rod embedded in a bilayer matrix," *Extreme Mechanics Letters*, vol. 25, pp. 1-6, Nov. 2018. DOI: 10.1016/J.EML.2018.10.004

Articles in Conference Proceedings

[C17] A. Frommer, **M. Abate**, P. Rivera-Ortiz, and Y. Diaz-Mercado, "A Capture Strategy for Multi-Pursuer Coordination Against a Fast Evader in 3D Reach-Avoid Games," *Modeling, Estimation and Control Conference (MECC)*, pp. 217-222, 2023. DOI: 10.1016/j.ifacol.2023.12.027.

[C16] A. Davydov, S. Jafarpour, **M. Abate**, F. Bullo, and S. Coogan, "Comparative Analysis of Interval Reachability for Robust Implicit and Feedforward Neural Networks," *2022 IEEE 61th Conference on Decision and Control (CDC)*, 2022. DOI: 10.1109/CDC51059.2022.9993217.

[C15] S. Jafarpour*, A. Davydov*, **M. Abate**, F. Bullo and S. Coogan,"Robust Training and Verification of Implicit Neural Networks: A Non-Euclidean Contractive Approach," *1st Workshop on Formal Verification of Machine Learning*, 2022. DOI: 10.48550/arXiv.2208.03889.

[C14] S. Jafarpour*, **M. Abate***, A. Davydov*, F. Bullo and S. Coogan, "Robustness Certificates for Implicit Neural Networks: A Mixed Monotone Contractive Approach," *4th Annual Learning for Dynamics & Control Conference (L4DC)*, 2022. **Selected for Oral Presentation—Top 20 percent of accepted papers**. Corpus ID: 245117816.

[C13] C. Llanes, **M. Abate** and S. Coogan, "Safety from Fast, In-the-Loop Reachability with Application to UAVs," *13th ACM / IEEE International Conference on Cyber-Physical Systems (ICCPS '22)*, pp. 127-136, 2022. DOI: 10.1109/ICCPS54341.2022.00018.

[C12] C. Llanes, **M. Abate** and S. Coogan, "Safety from In-the-Loop Reachability for Cyber-Physical Systems," *Workshop on Computation-Aware Algorithmic Design for Cyber-Physical Systems (CAADCPS '21)*, 2021. DOI: 10.1145/3457335.3461706.

[C11] **M. Abate**, C. Klett, S. Coogan and E. Feron, "Pointwise-in-Time Analysis and Non-Quadratic Lyapunov Functions for Linear Time-Varying Systems," *2021 American Control Conference (ACC)*, pp. 3550-3555, 2021. DOI: 10.23919/ACC50511.2021.9483291.

[C10] **M. Abate** and S. Coogan, "Improving the Fidelity of Mixed-Monotone Reachable Set Approximations via State Transformations," *2021 American Control Conference (ACC)*, pp. 4674-4679, 2021. DOI: 10.23919/ACC50511.2021.9483264.

[C9] G. Immanuel, **M. Abate** and Eric Feron, "Lyapunov Differential Equation Hierarchy and Polynomial Lyapunov Functions for Switched Implicit Systems," *2021 American Control Conference (ACC)*, pp. 2309-2314, 2021. DOI: 10.23919/ACC50511.2021.9482717.

[C8] C. Klett, **M. Abate**, S. Coogan and E. Feron, "A Numerical Method to Compute Stability Margins of Switching Linear Systems," *2021 American Control Conference (ACC)*, pp. 864-869, 2021. DOI: 10.23919/ACC50511.2021.9483141.

[C7] **M. Abate**, M. Mote, E. Feron and S. Coogan, "Verification and Runtime Assurance for Dynamical Systems with Uncertainty," *Proceedings of the 24th International Conference on Hybrid Systems: Computation and Control (HSCC)*, 2021. DOI: 10.1145/3447928.3456656.

[C6] **M. Abate** and S. Coogan, "Enforcing Safety at Runtime for Systems with Disturbances," *2020 IEEE 59th Conference on Decision and Control (CDC)*, pp. 2038-2043, 2020. DOI: 10.1109/CDC42340.2020.9304203.

[C5] **M. Abate** and S. Coogan, "Computing Robustly Forward Invariant Sets for Mixed-Monotone Systems," *2020 IEEE 59th Conference on Decision and Control (CDC)*, pp. 4553-4559, 2020. **Hybrid Systems TC Outstanding Student Paper Finalist**. DOI: 10.1109/CDC42340.2020.9304461.

[C4] **M. Abate**, C. Klett, S. Coogan and E. Feron, "Lyapunov Differential Equation Hierarchy and Polynomial Lyapunov Functions for Switched Linear Systems," *2020 American Control Conference (ACC)*, pp. 5322-5327, 2020. DOI: 10.23919/ACC45564.2020.9147662.

[C3] C. Klett, **M. Abate**, Y. Yoon, S. Coogan and E. Feron, "Bounding the State Covariance Matrix for Switched Linear Systems with Noise," *2020 American Control Conference (ACC)*, pp. 2876-2881, 2020. DOI: 10.23919/ACC45564.2020.9147787.

[C2] M. Dutreix, C. Santoyo, **M. Abate** and S. Coogan, "Interval-Valued Markov Chain Abstraction of Stochastic Systems Using Barrier Functions," *2020 American Control Conference (ACC)*, pp. 3583-3588, 2020. DOI: 10.23919/ACC45564.2020.9147744.

[C1] **M. Abate**, E. Feron and S. Coogan, "Monitor-Based Runtime Assurance for Temporal Logic Specifications," *2019 IEEE 58th Conference on Decision and Control (CDC)*, pp. 1997-2002, 2019. DOI: 10.1109/CDC40024.2019.9029716.

Other Publications, and Personal Projects

[O7] P. Mallory et al., "We Found the Best Shuffled Deck," 2024. Paper author. In Submission, Sigbovik 2024.

[O6] M. Abate, "Autograms," *The Journal of Wordplay*, Issue #4, pp. 54 - 59, 19 Feb., 2023 (read).

[O5] "How to Write a Paper," 2024 (pdf).

[O4] mattsmovielist.com: an Elo ranking game for movies and TV shows. Summer 2023 (site).

[O3] BeamerGen: A LaTex editor with an AI agent that knows your papers. Spring 2023 (demo, site).

[O2] Tango (2022): A Fugue in Film (watch).

[O1] "Gas-displacement Volumetry," inventor. US Patent: US10337902B2.

Invited Talks

[T2] "Efficient Reachability Analysis for Mixed Monotone Dynamical Systems," Invited Lecture, *Department of Mechanical Engineering, University of California Santa Barbara*, Virtual, Feb. 16 2022.

[T1] "Correctness with Complex Constraints: Runtime Assurance and Temporal Logics," Invited Lecture, *Air Force Research Laboratory, Controls Group*, Dayton, OH, Nov. 21 2019.

Mentoring

Jul. 2020 - Jun. 2021	Christian Llanes (Ph.D. Student, Robotics, Georgia Tech)
May 2020 - Dec. 2020	Christopher Lindbeck (B.S. Student, CS, Georgia Tech)
May 2020 - Dec. 2020	Thomas Contis (B.S. Student, ECE, Georgia Tech)
Jan. 2019 - Jan. 2020	Elizabeth Pruka (B.S. Student, ECE, Georgia Tech)
Aug. 2018 - Oct. 2020	William Stuckey (M.S. Student, ECE, Georgia Tech)

Teaching

Fall 2021	Teaching Practicum (Advisor: Nader Sadegh), Georgia Tech ME 6401, Linear Control Systems
Fall 2021	Guest Lecturer (Main Professor: Samuel Coogan), Georgia Tech ECE 3550, Feedback Control Systems
Spring 2021	Guest Lecturer (Main Professor: Samuel Coogan), Georgia Tech ECE 6552, Nonlinear Systems and Control

Honors, Nominations and Leadership

2022	Best Ph.D. Thesis Award, Nomination (Sigma Xi, Georgia Tech)
2022	Hybrid Systems TC Outstanding Student Paper Prize, Finalist (IEEE, Control Systems Society)
2017	McDonald-Smith Prize (Dartmouth College, Department of Music)
2016	Citation for Superior Performance as a Teaching Assistant (Thayer School of Engineering at Dartmouth College)
2015	Chapter President (Alpha Chi Alpha Fraternity at Dartmouth College)
2011	Eagle Scout (Boy Scouts of America)

References

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